

CloudPulse

Cloud Latency Monitoring & Performance Analysis

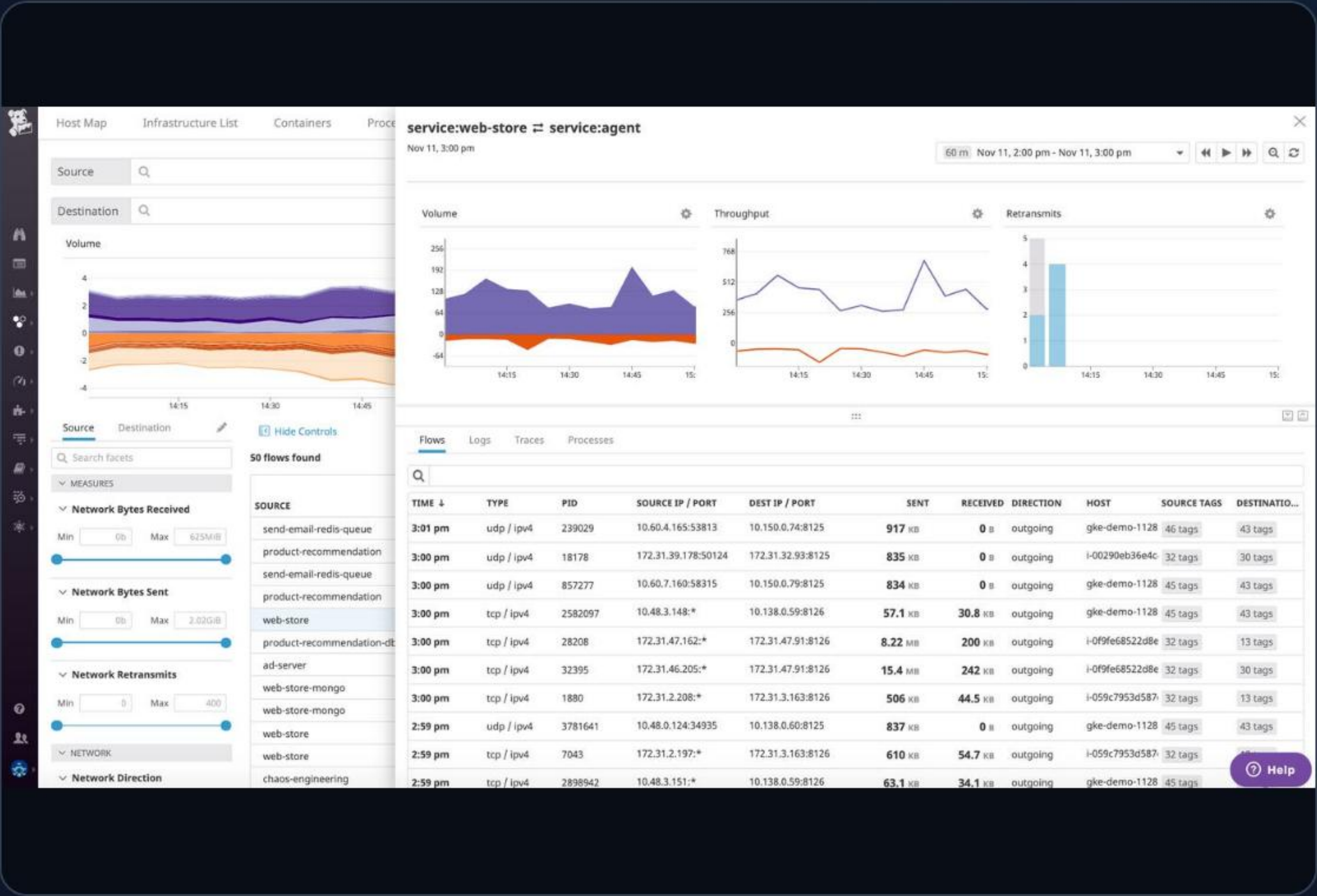
A modern, full-stack web application designed for real-time website latency testing, performance tracking, and comparative metrics visualization.

Core Architecture & Features

Engineered for high performance, reliability, and security in cloud
network observability.

Real-Time Latency & Feature Suite

-  **Instant Latency Testing:** Accurate measure of response time, HTTP status code, and payload size.
-  **Performance Dashboard:** Interactive historical trends analysis and latency pattern identification.
-  **Multi-URL Comparison:** Side-by-side performance benchmarking for up to 10 endpoints simultaneously.
-  **Enterprise Security:** Built-in SSRF protection, strict input verification, and sanitized error boundaries.



Modern 3-Tier Tech Stack

Layer	Technology Stack	Key Role & Capabilities
Frontend	React + Vite + Tailwind CSS	Responsive UI, rapid hot-reloading, utility-first styling
Backend	Node.js + Express REST API	Asynchronous latency service, controller logic & sanitization
Database	MongoDB Atlas	Persistent storage for historical test benchmarks and logs
Charts	Chart.js Integration	Interactive time-series line charts and comparative bar graphs

Interactive Analytics & Dashboard

Key Performance Statistics

Track essential metrics at a glance with live KPI cards displaying overall total tests executed, mean latency across regions, min/max response times, and system health status indicators.

Multi-URL Comparison Engine

Execute parallel requests across multiple web services. Automatically highlights fastest endpoints in green and slowest in red, generating rapid actionable competitive insight.

Enterprise Security & Protection

SSRF & Input Defense

CloudPulse implements multi-layered security controls to ensure safe remote URL monitoring without exposing internal infrastructure.

Features include strict blocking of localhost, private IP ranges (RFC 1918), and cloud metadata endpoints (169.254.169.254). Strict HTTP/HTTPS validation prevents protocol smuggling and injection attacks.

Security Diagram PowerPoint Template

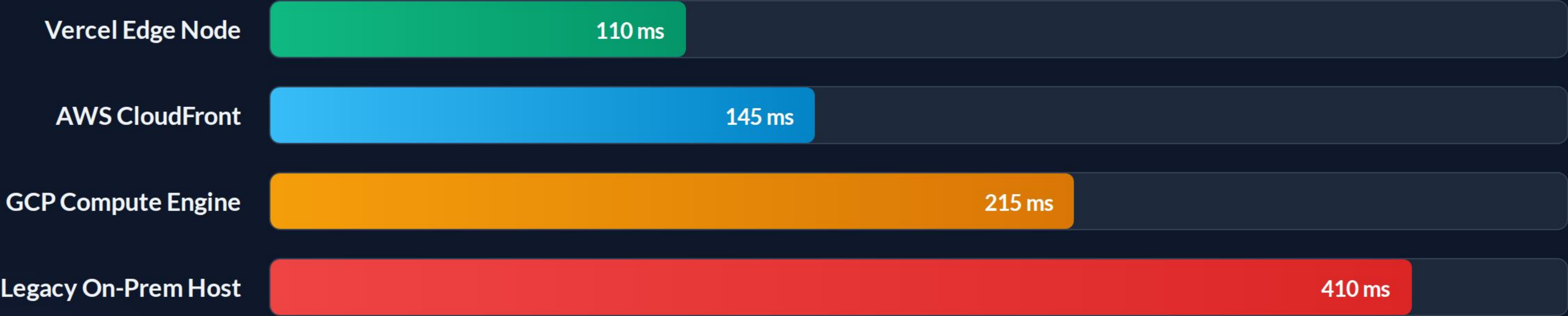
IDENTITIES

INFRASTRUCTURE



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Multi-URL Latency Benchmarks



Comparison highlights significant response time advantages of modern edge server distributions vs traditional hosting environments.

Robust Development & QA Process



Clean Architecture

Modular component structure with clear SOC (Separation of Concerns), reusable hooks, and sanitized REST controllers.



Rigorous QA

100% security test coverage against SSRF vectors, validated input sanitization, and full database persistence checks.



Accessibility

Full ARIA compliance, structured keyboard navigation support, and WCAG 2.1 Level AA ready color contrast.

Key Performance Highlights

<2s

Page Initial Load Time

<500ms

API Latency Test Response Time

<100ms

Chart Render & Update Time

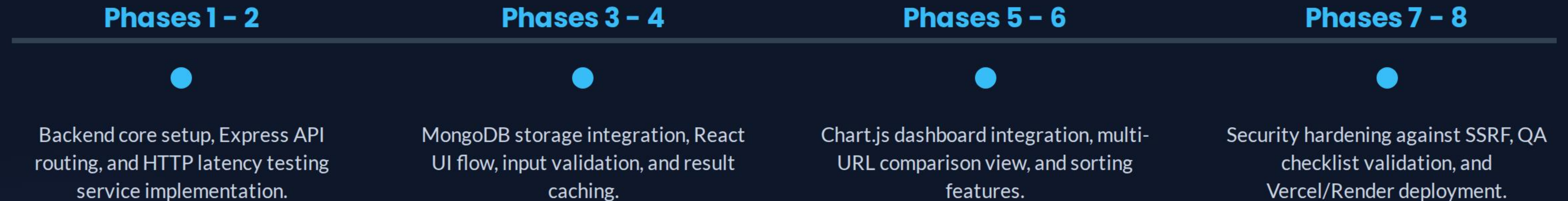
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Storage per Test Record

100%

SSRF Protection Validation

Development Phase Roadmap



Production-Ready Cloud Deployment

Deployed entirely on zero-cost free-tier infrastructure with high availability and automated CI/CD pipelines.

MongoDB Atlas

512MB document storage,
automated security rules.

Render Web Service

Node.js backend, auto-scaling,
built-in HTTPS.

Vercel Edge

Global CDN distribution for
React client interface.

Questions & Answers

Thank you for exploring CloudPulse Latency Monitoring Solution.

 Production Ready | Open Source Performance Monitoring

Image Sources



<https://web-assets.dd-static.net/42588/1776292330-cloud-network-monitoring-network-performance-monitoring-logs-3-new.png?format=auto&fit=bounds&quality=75&disable=upscale&width=1400&dpr=1>

Source: www.datadoghq.com



<https://slidemodel.com/wp-content/uploads/23181-02-6-item-cyber-security-diagram-powerpoint-template-16x9-1.jpg>

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